K.L.E Society's

College of Bachelor of Computer Applications (BCA)

Gokak (4419)

(Affiliated to Rani Channamma University, Belagavi.)

CERTIFICATE

2024-25

This is to certify that Mr/Miss_____

_____

has satisfactorily completed the _____ lab in B.C.A I Semester as prescribed by the Rani Channamma University Belagavi, under our supervision during the period 2024-25.

Roll No._____ Reg.No(University)._____

Staff Incharge H.O.D

Examiners

1._____

2._____

INDEX

Sl.No	Topic	Page	Remarks
1.	Case Study 1: (Select clause, Arithmetic Operators) Database: Employee	1 -9	
2.	Case Study 2: (group by, having clause) Database: Bank system	10-16	
3.	Case Study 3: (Logical, Relational Operators) Database: Library	17-22	
4.	Case Study 4 Database: Subject	23-27	

Case Study 1: (Select clause, Arithmetic Operators)

Database: Employee

1. Create the following tables with suitable constraints and insert tuples.

Table: Employee

Emp_Id	First_Name	Last_Name	Hire_Date	Address	City
1001	Raghav	Patel	11-May-06	83 first street	Belagavi
1002	Mohini	Rathi	25-Feb-08	842 Vine Ave	Athani
1012	Sham	Pujari	12-Sep-05	33 Elm St.	Mysore
1015	Kriti	Dev	19-Dec-06	11 Red Road	Belagavi
1016	Sarath	Sharma	22-Aug-07	440 MG Road	New Delhi
1020	Monika	Gupta	07-Jun-08	9 Bandra	Mumbai
1021	Rohit	Sharma	06-jun-08	Hill road	Paris

Table : EmpSalary

Emp_Id	Salary	Benefits	Designation
1001	10000	3000	Manager
1002	8000	1200	Salesman
1012	20000	5000	Director
1015	6500	1300	Clerk
1016	6000	1000	Clerk
1020	8000	1200	Salesman
1021	8000	1200	Salesman

Write queries for the following

1. Display FIRSTNAME, LASTNAME, ADDRESS, and CITY of all employees living in PARIS.
2. Display the content of the employee table in descending order of FIRSTNAME
3. Select the FIRSTNAME and SALARY of the salesman
4. To display the FIRSTNAME, LASTNAME, AND TOTAL SALARY of all employees from the table EMPLOYEE and EMPSALARY. Where TOTAL SALARY is calculated as SALARY+BENEFITS
5. List the Names of employees, who are more than 1 year old in the organization
6. Count the number of distinct DESIGNATION from EMPSALARY
7. List the employees whose names have exactly 6 characters
8. Add new column PHONE_NO to EMPLOYEE and update the records
9. List employee names, who have joined before 15-Jun-08 and after 16-Jun-07
10. Generate Salary slip with Name, Salary, Benefits, HRA-50%, DA-30%, PF-12%, Calculate gross. Order the result in descending order of the gross.

Open SQL Command Prompt

SQL> conn

Enter user-name: system
Enter password: manager

Connected.

SQL> create table Employee(Emp_Id int primary key,First_Name varchar(20),Last_Name varchar(20),Hire_Date date,Address varchar(50),City varchar(20));

Table created.

SQL> desc Employee

Name	Null?	Type
EMP_ID	NOT NULL	NUMBER(38)
FIRST_NAME		VARCHAR2(20)
LAST_NAME		VARCHAR2(20)
HIRE_DATE		DATE
ADDRESS		VARCHAR2(50)
CITY		VARCHAR2(20)

SQL> create table EmpSalary(Emp_Id int primary key,Salary int,Benefits int,Designation varchar(20), foreign key(Emp_Id)references Employee(Emp_Id));

Table created.

SQL> desc EmpSalary

Name	Null?	Type
EMP_ID	NOT NULL	NUMBER(38)
SALARY		NUMBER(38)
BENEFITS		NUMBER(38)
DESIGNATION		VARCHAR2(20)

Insert into Employee Table

SQL> insert into Employee values('1001','Raghav','Patel','11-May-2006','83 first street','Belagavi');

1 row created.

SQL> insert into Employee values('1002','Mohini','Rathi','25-Feb-2008','842 Vins Ave','Athani');

1 row created.

SQL> insert into Employee values('1012','Sham','Pujari','12-Sep-2005','33 Elm St.','Mysore');

1 row created.

SQL> insert into Employee values('1015','Kirti','Dev','19-Dec-2006','11 Red Road','Belagavi');

1 row created.

SQL> insert into Employee values('1016','Sarath','Sharma','22-Aug-2007','440 MG Road','New Delhi');

1 row created.

SQL> insert into Employee values('1020','Monika','Gupta','07-Jun-2008','9 Bandra','Mumbai');

1 row created.

SQL> insert into Employee values('1021','Rohit','Sharma','08-Jun-2008','Hill Road','Paris');

1 row created.

SQL> Select * from Employee;

EMP_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	ADDRESS	CITY
1001	Raghav	Patel	11-MAY-06	83 first street	Belagavi
1002	Mohini	Rathi	25-FEB-08	842 Vins Ave	Athani
1012	Sham	Pujari	12-SEP-05	33 Elm St.	Mysore
1015	Kirti	Dev	19-DEC-06	11 Red Road	Belagavi
1016	Sarath	Sharma	22-AUG-07	440 MG Road	New Delhi
1020	Monika	Gupta	07-JUN-08	9 Bandra	Mumbai
1021	Rohit	Sharma	08-JUN-08	Hill Road	Paris

7 rows selected.

Insert into EmpSalary Table

```
SQL> insert into EmpSalary values('1001','10000','3000','Manager');

1 row created.

SQL> insert into EmpSalary values('1002','8000','1200','Salesman');

1 row created.

SQL> insert into EmpSalary values('1012','20000','5000','Director');

1 row created.

SQL> insert into EmpSalary values('1015','6500','1300','Clerk');

1 row created.

SQL> insert into EmpSalary values('1016','6000','1000','Clerk');

1 row created.

SQL> insert into EmpSalary values('1020','8000','1200','Salesman');

1 row created.

SQL> insert into EmpSalary values('1021','8000','1200','Salesman');

1 row created.

SQL> select * from EmpSalary;

   EMP_ID   SALARY   BENEFITS DESIGNATION
---------- ---------- ---------- ------------------------------
     1001      10000       3000    Manager
     1002       8000       1200    Salesman
     1012      20000       5000    Director
     1015       6500       1300    Clerk
     1016       6000       1000    Clerk
     1020       8000       1200    Salesman
     1021       8000       1200    Salesman
7 rows selected.
SQL> commit;

Commit complete.
```

Query

1. Display FIRSTNAME, LASTNAME, ADDRESS, and CITY of all employees living in PARIS.

```
SQL> select First_Name,Last_Name,Address,City
  2  from Employee
  3  where City='Paris';
```

```
FIRST_NAME   LAST_NAME      ADDRESS       CITY
-------------------------------------------------- --------------------
  Rohit             Sharma           Hill Road        Paris
```

2. Display the content of the employee table in descending order of FIRSTNAME

```
SQL> select * from Employee
  2  order by First_Name DESC;
```

```
   EMP_ID     FIRST_NAME      LAST_NAME     HIRE_DATE            ADDRESS         CITY
---------------------------------------------- ---------------------------------------------------------------------
     1012        Sham             Pujari          12-SEP-05          33 Elm St.         Mysore
     1016        Sarath           Sharma          22-AUG-07          440 MG Road        New Delhi
     1021        Rohit            Sharma          08-JUN-08          Hill Road          Paris
     1001        Raghav           Patel           11-MAY-06          83 first street    Belagavi
     1020        Monika           Gupta           07-JUN-08          9 Bandra           Mumbai
     1002        Mohini           Rathi           25-FEB-08          842 Vins Ave       Athani
     1015        Kirti            Dev             19-DEC-06          11 Red Road        Belagavi
```

7 rows selected.

3. Select the FIRSTNAME and SALARY of the salesman

```
SQL> select First_Name,Salary
  2  from Employee,EmpSalary
  3  where Employee.Emp_Id=EmpSalary.Emp_Id AND Designation='Salesman';
```

```
FIRST_NAME           SALARY
-----------------------------------------
Mohini                8000
Monika                8000
Rohit                 8000
```

4. To display the FIRSTNAME, LASTNAME, AND TOTAL SALARY of all employees from the table EMPLOYEE and EMPSALARY. Where TOTAL SALARY is calculated as SALARY+BENEFITS

```
SQL> select First_Name,Last_Name,(Salary+Benefits) AS Total_Salary
  2  from Employee,EmpSalary
  3  where Employee.Emp_Id=EmpSalary.Emp_Id;

FIRST_NAME   LAST_NAME       TOTAL_SALARY
-------------------- -------------------- ------------------------
Raghav            Patel                 13000
Mohini            Rathi                 9200
Sham              Pujari                25000
Kirti             Dev                   7800
Sarath            Sharma                7000
Monika            Gupta                 9200
Rohit             Sharma                9200

7 rows selected.
```

5. List the Names of employees, who are more than 1 year old in the organization

```
SQL> select First_Name,Last_Name
  2  from Employee
  3  where (months_between(sysdate,Hire_Date)/12)>=1;

FIRST_NAME   LAST_NAME
----------------------------------------
Raghav            Patel
Mohini            Rathi
Sham              Pujari
Kirti             Dev
Sarath            Sharma
Monika            Gupta
Rohit             Sharma

7 rows selected.
```

6. Count the number of distinct DESIGNATION from EMPSALARY

```
SQL> select count(distinct Designation)
  2  from EmpSalary;

COUNT(DISTINCTDESIGNATION)
--------------------------------------------
                         4
```

7. List the employees whose names have exactly 6 characters

```
SQL> select * from Employee
  2  where First_Name LIKE '_____';
```

(6 underscores(_) need to add)

EMP_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	ADDRESS	CITY
1001	Raghav	Patel	11-MAY-06	83 first street	Belagavi
1002	Mohini	Rathi	25-FEB-08	842 Vins Ave	Athani
1016	Sarath	Sharma	22-AUG-07	440 MG Road	New Delhi
1020	Monika	Gupta	07-JUN-08	9 Bandra	Mumbai

4 rows selected.

8. Add new column PHONE_NO to EMPLOYEE and update the records

```
SQL> alter table Employee add Phone_No number(10);

Table altered.

SQL> Update Employee set Phone_No=123456789 where Emp_Id='1001';

1 row updated.

SQL> Update Employee set Phone_No=123456788 where Emp_Id='1002';
```

1 row updated.

SQL> Update Employee set Phone_No=123456787 where Emp_Id='1012';

1 row updated.

SQL> Update Employee set Phone_No=123456786 where Emp_Id='1015';

1 row updated.

SQL> Update Employee set Phone_No=123456785 where Emp_Id='1016';

1 row updated.

SQL> Update Employee set Phone_No=123456784 where Emp_Id='1020';

1 row updated.

SQL> Update Employee set Phone_No=123456783 where Emp_Id='1021';

1 row updated.

EMP_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	ADDRESS	CITY	Phone_No
1001	Raghav	Patel	11-MAY-06	83 first street	Belagavi	123456789
1002	Mohini	Rathi	25-FEB-08	842 Vins Ave	Athani	123456788
1012	Sham	Pujari	12-SEP-05	33 Elm St.	Mysore	123456787
1015	Kirti	Dev	19-DEC-06	11 Red Road	Belagavi	123456786
1016	Sarath	Sharma	22-AUG-07	440 MG Road	New Delhi	123456785
1020	Monika	Gupta	07-JUN-08	9 Bandra	Mumbai	123456784
1021	Rohit	Sharma	08-JUN-08	Hill Road	Paris	123456783

7 rows selected.

9. List employee names, who have joined before 15-Jun-08 and after 16-Jun-07

```
SQL> select First_Name,Last_Name
  2  from Employee
  3  where Hire_Date between '16-jun-2007' and '15-jun-2008';

FIRST_NAME     LAST_NAME
-------------------- --------------------
Mohini               Rathi
Sarath               Sharma
Monika               Gupta
Rohit                Sharma
```

10. Generate Salary slip with Name, Salary, Benefits, HRA-50%, DA-30%, PF-12%, Calculate gross. Order the result in descending order of the gross.

```
SQL> select First_Name, Salary,Benefits,
2 Salary*0.5 as HRA,
3 Salary*0.3 as DA,
4 Salary*0.12 as PF,
5 Salary+Salary*0.5+Salary*0.3-Salary*0.12 as Gross_Salary
6 from Employee,EmpSalary
7 where Employee.Emp_Id=EmpSalary.Emp_Id
 8 order by Gross_Salary desc;

FIRST_NAME     SALARY    BENEFITS       HRA        DA        PF      GROSS_SALARY
-------------------------------------------------------------------------------------------------------
Sham              20000       5000          10000       6000     2400       33600
Raghav            10000       3000          5000        3000     1200       16800
Mohini            8000        1200          4000        2400     960        13440
Rohit             8000        1200          4000        2400     960        13440
Monika            8000        1200          4000        2400     960        13440
Kirti             6500        1300          3250        1950     780        10920
Sarath            6000        1000          3000        1800     720        10080

7 rows selected.
```

SQL>exit;

Case Study 2: (group by, having clause) Database: Bank system
Create the following table and insert tuples with suitable constraints

Table: Account

Account_No	Cust_Name	Branch_ID
AE0012856	Reena	SB002
AE1185698	Akhil	SB001
AE1203996	Daniel	SB004
AE1225889	Roy	SB002
AE8532166	Sowparnika	SB003
AE8552266	Anil	SB003
AE1003996	Saathwik	SB004
AE1100996	Swarna	SB002

Table: Branch

Branch_ID	Branch_Name	Branch_City
SB001	Malleshwaram	Bangalore
SB002	MG Road	Bangalroe
SB003	MG Road	Mysore
SB004	Jainagar	Mysore

Table: Depositor

Account_No	Branch_ID	Balance
AE0012856	SB002	12000
AE1203996	SB004	58900
AE8532166	SB003	40000
AE1225889	SB002	150000

Table: Loan

Account_No	Branch_ID	Balance
AE1185698	SB001	102000
AE8552266	SB003	40000
AE1003996	SB004	15000
AE1100996	SB002	100000

1. Display the Total Number of accounts present in each branch.
2. Display the Total Loan amount in each branch.
3. Display the Total deposited amount in each branch by descending order.
4. Display max , min loan amount present in each city.
5. Display average amount deposited in each branch, each city.
6. Display maximum of loan amount in each branch where the balance is more than 25000.
7. Display Total Number of accounts present in each city.
8. Display all customer details in ascending order of branch_id.
9. Update Balance to 26000 where account_no=AE1203996
10. Display Customer Names with their branch Name

Table Creation

SQL> create table Branch(Branch_ID varchar(10) primary key,Branch_Name varchar(20),Branch_City varchar(20));

Table created.

SQL> create table Account(Account_No varchar(20) primary key,Cust_Name varchar(20),Branch_ID varchar(10)references Branch(Branch_ID));

Table created.

SQL> create table Depositor(Account_No varchar(20)references Account(Account_No),Branch_ID varchar(20)references Branch(Branch_ID),Balance int);

Table created.

SQL> create table Loan(Account_No varchar(20)references Account(Account_No),Branch_ID varchar(20)references Branch(Branch_ID),Balance int);

Table created.

SQL> INSERT INTO Branch VALUES('SB001','Malleshwaram','Bangalore');

1 row created.

SQL> INSERT INTO Branch VALUES('SB002', 'MG Road', 'Bangalore');

1 row created.

SQL> INSERT INTO Branch VALUES('SB003', 'MG Road', 'Mysore');

1 row created.

SQL> INSERT INTO Branch VALUES('SB004', 'Jainagar', 'Mysore');

1 row created.

SQL> select * from Branch;

BRANCH_ID	BRANCH_NAME	BRANCH_CITY
SB001	Malleshwaram	Bangalore
SB002	MG Road	Bangalore
SB003	MG Road	Mysore
SB004	Jainagar	Mysore

```
SQL> INSERT INTO Account VALUES('AE0012856', 'Reena', 'SB002');
1 row created.

SQL> INSERT INTO Account VALUES('AE1185698', 'Akhil', 'SB001');
1 row created.

SQL> INSERT INTO Account VALUES('AE1203996', 'Daniel', 'SB004');
1 row created.

SQL> INSERT INTO Account VALUES('AE1225889', 'Roy', 'SB002');
1 row created.

SQL> INSERT INTO Account VALUES('AE8532166', 'Sowparnika', 'SB003');
1 row created.

SQL> INSERT INTO Account VALUES('AE8552266', 'Anil', 'SB003');
1 row created.

SQL> INSERT INTO Account VALUES('AE1003996', 'Saathwik', 'SB004');
1 row created.

SQL> INSERT INTO Account VALUES('AE1100996', 'Swarna', 'SB002');
1 row created.
```

SQL> select * from Account;

ACCOUNT_NO	CUST_NAME	BRANCH_ID
AE0012856	Reena	SB002
AE1185698	Akhil	SB001
AE1203996	Daniel	SB004
AE1225889	Roy	SB002
AE8532166	Sowparnika	SB003
AE8552266	Anil	SB003
AE1003996	Saathwik	SB004
AE1100996	Swarna	SB002

```
SQL> INSERT INTO Depositor VALUES('AE0012856', 'SB002', 12000);
1 row created.

SQL> INSERT INTO Depositor VALUES('AE1203996', 'SB004', 58900);
1 row created.

SQL> INSERT INTO Depositor VALUES('AE8532166', 'SB003', 40000);
1 row created.

SQL>  INSERT INTO Depositor VALUES('AE1225889', 'SB002', 150000);
1 row created.
```

SQL> select * from Depositor;

```
ACCOUNT_NO         BRANCH_ID       BALANCE
------------------------------------------------------------------------
AE0012856          SB002           12000
AE1203996          SB004           58900
AE8532166          SB003           40000
AE1225889          SB002           150000
```

```
SQL> INSERT INTO Loan VALUES ('AE1185698', 'SB001', 102000);
1 row created.

SQL> INSERT INTO Loan VALUES('AE8552266', 'SB003', 40000);
1 row created.

SQL> INSERT INTO Loan VALUES('AE1003996', 'SB004', 15000);
1 row created.
```

SQL> select * from Loan;

```
ACCOUNT_NO         BRANCH_ID          BALANCE
-------------------------------------------------------------------------------
AE1185698          SB001              102000
AE8552266          SB003              40000
AE1003996          SB004              15000
```

SQL> commit;

Commit complete.

1. Display the Total Number of accounts present in each branch.

```
SQL> SELECT Branch_ID, COUNT(*) AS Total_Accounts
  2  FROM Account GROUP BY Branch_ID;

BRANCH_ID  TOTAL_ACCOUNTS
-------------------------------------------------
SB004              2
SB001              1
SB002              3
SB003              2
```

2. Display the Total Loan amount in each branch

```
SQL> SELECT Branch_ID, SUM(Balance) AS Total_Loan_Amount
  2  FROM Loan GROUP BY Branch_ID;

BRANCH_ID          TOTAL_LOAN_AMOUNT
------------------- ----------------------------------------
SB004                       15000
SB001                      102000
SB003                       40000
```

3. Display the Total deposited amount in each branch by descending order.

```
SQL> SELECT Branch_ID, SUM(Balance) AS Total_Deposited_Amount
  2  FROM Depositor GROUP BY Branch_ID
  3  ORDER BY Total_Deposited_Amount DESC;

BRANCH_ID          TOTAL_DEPOSITED_AMOUNT
------------------- ----------------------------------------------
SB002                      162000
SB004                       58900
SB003                       40000
```

4. Display max , min loan amount present in each city.

```
SQL> SELECT Branch_City, MAX(L.Balance),MIN(L.Balance)
  2  From Loan L,Branch B
  3  where L.Branch_ID=B.Branch_ID
  4  group by Branch_City;

BRANCH_CITY        MAX(L.BALANCE)          MIN(L.BALANCE)
------------------- -------------- ----------------------------------------------------
Mysore                   40000                       15000
Bangalore               102000                      102000
```

5.**Display average amount deposited in each branch, each city.**

```
SQL> SELECT B.Branch_ID, B.Branch_City, AVG(D.Balance) AS Avg_Deposit
  2  FROM Depositor D, Branch B
  3  Where D.Branch_ID = B.Branch_ID
  4  GROUP BY B.Branch_ID, B.Branch_City;

BRANCH_ID  BRANCH_CITY        AVG_DEPOSIT
-------------------------------------------------------------------------
SB004          Mysore              58900
SB002          Bangalore           81000
SB003          Mysore              40000
```

6. Display maximum of loan amount in each branch where the balance is more than 25000.

```
SQL> SELECT Branch_ID, MAX(Balance) AS Max_Loan
  2  FROM Loan
  3  WHERE Balance > 25000
  4  GROUP BY Branch_ID;

BRANCH_ID          MAX_LOAN
-----------------------------------------------
SB001                102000
SB003                40000
```

7. Display Total Number of accounts present in each city.

```
SQL> SELECT B.Branch_City, COUNT(A.Account_No) AS Total_Accounts
  2  FROM Account A,Branch B
  3  WHERE A.Branch_ID = B.Branch_ID
  4  GROUP BY B.Branch_City;

BRANCH_CITY        TOTAL_ACCOUNTS
-------------------------------------------------------------
Mysore                4
Bangalore             4
```

8. Display all customer details in ascending order of branch_id.

```
SQL> SELECT * FROM Account
  2   ORDER BY Branch_ID ASC;

ACCOUNT_NO   CUST_NAME         BRANCH_ID
---------- ----------------------------------------------------
AE1185698        Akhil                  SB001
AE0012856        Reena                  SB002
AE1225889        Roy                    SB002
AE1100996        Swarna                 SB002
AE8532166        Sowparnika             SB003
AE8552266        Anil                   SB003
AE1003996        Saathwik               SB004
AE1203996        Daniel                 SB004
```

9. Update Balance to 26000 where account_no=AE1203996.

```
SQL> UPDATE Depositor
  2  SET Balance = 26000
  3  WHERE Account_No = 'AE1203996';
1 row updated.

SQL> select * from Depositor;

ACCOUNT_NO         BRANCH_ID       BALANCE
------------------------------------------------------------------------
AE0012856              SB002            12000
AE1203996              SB004            26000
AE8532166              SB003            40000
AE1225889              SB002            150000
```

10. Display Customer Names with their branch Name

```
SQL> SELECT A.Cust_Name, B.Branch_Name
  2  FROM Account A,Branch B
  3  WHERE A.Branch_ID = B.Branch_ID;

CUST_NAME          BRANCH_NAME
-----------------------------------------------------
Reena                  MG Road
Akhil                  Malleshwaram
Daniel                 Jainagar
Roy                    MG Road
Sowparnika             MG Road
Anil                   MG Road
Saathwik               Jainagar
Swarna                 MG Road

8 rows selected.
```

Case Study 3: (Logical, Relational Operators)
Database: Library
Create Following tables and insert tuples with suitable constraints

Table: Books

Book_Id	Book_Name	Author_Name	Publishers	Price	Type	Quantity
C0001	The Klone and I	Lata Kappor	EPP	355	Novel	5
F0001	The Tears	William Hopkins	First Publ	650	Fiction	20
T0001	My First C++	Brain & Brooke	ERP	350	Text	10
T0002	C++ Brainwork"s	A.W.Rossaine	TDH	350	Text	15
F0002	Thunderbolts	Ana Roberts	First Publ.	750	Fiction	50

Table: Issued

Book_Id	Quantity_Issued
T0001	4
C0001	5
F0001	2
T0002	5
F0002	8

Write queries for the following

1. To show Book name, Author name and price of books of First Publ. publisher
2. Display Book id, Book name and publisher of books having quantity more than 8 and price less than 500
3. Select Book id, book name, author name of books which is published by other than ERP publishers and price between 300 to 700
4. Generate a Bill with Book_id, Book_name, Publisher, Price, Quantity, 4% of VAT "Total"
5. Display book details with book id"s C0001, F0001, T0002, F0002 (Hint: use IN operator)
6. Display Book list other than, type Novel and Fiction
7. Display book details with the author name starting with letter "A"
8. Display book details with the author name starting with letter "T" and ending with "S"
9. Select Book_Id, Book_Name, Author Name, Quantity Issued where Books.Books_Id = Issued.Book_Id
10. List the book_name, Author_name, Price. In ascending order of Book_name and then in descending order of price

Table Creations

SQL> create table Books (Book_Id varchar(10) primary key, Book_Name varchar(50),Author_Name varchar(50),Publishers varchar(50),Price int,Type varchar(20),Quantity int);

Table created.

SQL> create table Issued (Book_Id varchar(5),Quantity_Issued int,foreign key (Book_Id) references Books(Book_Id));

Table created.

Insertions (Books Table)

SQL> insert into Books values('C0001','The Klone and I','Lata Kappor','EPP',355,'Novel',5);
1 row created.

SQL> insert into Books values('F0001', 'The Tears', 'William Hopkins', 'First Publ', 650, 'Fiction', 20);
1 row created.

SQL> insert into Books values('T0001', 'My First C++', 'Brain and Brooke', 'ERP', 350, 'Text', 10);
1 row created.

SQL> insert into Books values('T0002', 'C++ Brainwork', 'A.W.Rossaine', 'TDH', 350, 'Text', 15);
1 row created.

SQL> insert into Books values('F0002', 'Thunderbolts', 'Ana Roberts', 'First Publ.', 750, 'Fiction', 50);
1 row created.

SQL> select * from Books;

Book_Id	Book_Name	Author_Name	Publishers	Price	Type	Quantity
C0001	The Klone and I	Lata Kappor	EPP	355	Novel	5
F0001	The Tears	William Hopkins	First Publ	650	Fiction	20
T0001	My First C++	Brain & Brooke	ERP	350	Text	10
T0002	C++ Brainwork"s	A.W.Rossaine	TDH	350	Text	15
F0002	Thunderbolts	Ana Roberts	First Publ.	750	Fiction	50

Insertions (Issued Table)

SQL> insert into Issued values('T0001', 4);
1 row created.

SQL> insert into Issued values ('C0001', 5);
1 row created.

SQL> insert into Issued values ('F0001', 2);
1 row created.

```
SQL> insert into Issued values ('T0002', 5);
1 row created.

SQL> insert into Issued values ('F0002', 8);
1 row created.
```

SQL> select * from Issued;

```
BOOK_ID    QUANTITY_ISSUED
-----------------------------------------
T0001              4
C0001              5
F0001              2
T0002              5
F0002              8
```

SQL> commit;
Commit complete.

1. To show Book name, Author name and price of books of First Publ. publisher

```
SQL> SELECT Book_Name, Author_Name, Price
  2  FROM Books
  3  WHERE Publishers = 'First Publ';

BOOK_NAME    AUTHOR_NAME          PRICE
--------------------------------------------------------------------
The Tears          William Hopkins           650
```

2. Display Book id, Book name and publisher of books having quantity more than 8 and price less than 500

```
SQL> SELECT Book_Id, Book_Name, Publishers
  2  FROM Books
  3  WHERE Quantity > 8 AND Price < 500;

BOOK_ID    BOOK_NAME     PUBLISHERS
-----------------------------------------------------------
T0001         My First C++              ERP

T0002       C++ Brainwork             TDH
```

3. Select Book id, book name, author name of books which is published by other than ERP publishers and price between 300 to 700

```
SQL> SELECT Book_Id, Book_Name, Author_Name
  2  FROM Books
  3  WHERE Publishers!='ERP' AND Price BETWEEN 300 AND 700;
```

BOOK_ID	BOOK_NAME	AUTHOR_NAME
C0001	The Klone and I	Lata Kappor
F0001	The Tears	William Hopkins
T0002	C++ Brainwork	A.W.Rossaine

4. Generate a Bill with Book_id, Book_name, Publisher, Price, Quantity, 4% of VAT "Total"

```
SQL> SELECT Book_Id, Book_Name, Publishers, Price, Quantity,
  2  (Price * Quantity) * 0.04 AS VAT,
  3  (Price * Quantity) + (Price * Quantity) * 0.04 AS Total
  4  FROM Books;
```

BOOK_ID	BOOK_NAME	PUBLISHERS	PRICE	QUANTITY	VAT	TOTAL
C0001	The Klone and I	EPP	355	5	71	1846
F0001	The Tears	First Publ	650	20	520	13520
T0001	My First C++	ERP	350	10	140	3640
T0002	C++ Brainwork	TDH	350	15	210	5460
F0002	Thunderbolts	First Publ.	750	50	1500	39000

5. Display book details with book id"s C0001, F0001, T0002, F0002 (Hint: use IN operator)

```
SQL> SELECT *
  2  FROM Books
  3  WHERE Book_Id IN ('C0001', 'F0001', 'T0002', 'F0002');
```

BOOK_ID	BOOK_NAME	AUTHOR_NAME	PUBLISHERS	PRICE	TYPE	QUANTITY
C0001	The Klone and I	Lata Kappor	EPP	355	Novel	5
F0001	The Tears	William Hopkins	First Publ	650	Fiction	20
F0002	Thunderbolts	Ana Roberts	First Publ.	750	Fiction	50
T0002	C++ Brainwork	A.W.Rossaine	TDH	350	Text	15

6. Display Book list other than, type Novel and Fiction

```
SQL> SELECT *
  2  FROM Books
  3  WHERE Type NOT IN ('Novel', 'Fiction');
```

BOOK_ID	BOOK_NAME	AUTHOR_NAME	PUBLISHERS	PRICE	TYPE	QUANTITY
T0001	My First C++	Brain and Brooke	ERP	350	Text	10
T0002	C++ Brainwork	A.W.Rossaine	TDH	350	Text	15

7. Display book details with the author name starting with letter "A"

```
SQL> SELECT *
  2  FROM Books
  3  WHERE Author_Name LIKE 'A%';
```

BOOK_ID	BOOK_NAME	AUTHOR_NAME	PUBLISHERS	PRICE	TYPE	QUANTITY
T0002	C++ Brainwork	A.W.Rossaine	TDH	350	Text	15
F0002	Thunderbolts	Ana Roberts	First Publ.	750	Fiction	50

8. Display book details with the author name starting with letter "T" and ending with "S"

```
SQL> SELECT *
  2  FROM Books
  3  WHERE Author_Name LIKE 'T%S';

no rows selected
```

9. Select Book_Id, Book_Name, Author Name, Quantity Issued where Books.Books_Id = Issued.Book_Id

```
SQL> SELECT B.Book_Id, B.Book_Name, B.Author_Name, I.Quantity_Issued
  2  FROM Books B,Issued I
  3  where B.Book_Id = I.Book_Id;
```

BOOK_ID	BOOK_NAME	AUTHOR_NAME	QUANTITY_ISSUED
T0001	My First C++	Brain and Brooke	4
C0001	The Klone and I	Lata Kappor	5
F0001	The Tears	William Hopkins	2
T0002	C++ Brainwork	A.W.Rossaine	5
F0002	Thunderbolts	Ana Roberts	8

10. List the book_name, Author_name, Price. In ascending order of Book_name and then in descending order of price

```
SQL> SELECT Book_Name, Author_Name, Price
  2  FROM Books
  3  ORDER BY Book_Name ASC, Price DESC;
```

BOOK_NAME	AUTHOR_NAME	PRICE
C++ Brainwork	A.W.Rossaine	350
My First C++	Brain and Brooke	350
The Klone and I	Lata Kappor	355
The Tears	William Hopkins	650
Thunderbolts	Ana Roberts	750

Case Study 4
Database: Subject
Create Following table and insert tuples with suitable constraints

Table - Physics

Regno	Name	Year	Combination
AJ00325	Ashwin	First	PCM
AJ00225	Swaroop	Second	PMCs
AJ00385	Sarika	Third	PME
AJ00388	Rupa	First	PMCs

Table – Computer Science

Regno	Name	Year	Combination
AJ00225	Swaroop	Second	PMCs
AJ00296	Tejas	Second	BCA
AJ00112	Geeta	First	BCA
AJ00388	Rupa	First	PMCs

Write queries for the following-
1. Select all students from Physics and Computer Science
2. Select students common in Physics and Computer Science
3. Display all student details those are studying in the second year
4. Display students who are studying both Physics and computer science in the second year
5. Display the students studying only physics
6. Display the students studying only Computer Science
7. Select all students having PMCs combination
8. Select all students having a BCA combination
9. Select all students studying in the Third year
10. Rename table Computer Science to CS

Creating Tables

SQL> CREATE TABLE Physics(Regno VARCHAR(10) PRIMARY KEY,Name VARCHAR(50),Year VARCHAR(10),Combination VARCHAR(10));

Table created.

SQL> CREATE TABLE Computer_Science(Regno VARCHAR(10)PRIMARY KEY,Name VARCHAR(50),Year VARCHAR(10),Combination VARCHAR(10));

Table created.

Inserting a Values in Physics Table

SQL> INSERT INTO Physics VALUES('AJ00325', 'Ashwin', 'First', 'PCM');
1 row created.

SQL> INSERT INTO Physics VALUES ('AJ00225', 'Swaroop', 'Second', 'PMCs');
1 row created.

SQL> INSERT INTO Physics VALUES ('AJ00385', 'Sarika', 'Third', 'PME');
1 row created.

SQL> INSERT INTO Physics VALUES ('AJ00388', 'Rupa', 'First', 'PMCs');
1 row created.

SQL> select * from Physics;

REGNO	NAME	YEAR	COMBINATION
AJ00325	Ashwin	First	PCM
AJ00225	Swaroop	Second	PMCs
AJ00385	Sarika	Third	PME
AJ00388	Rupa	First	PMCs

Inserting a Values in Computer_Science Table-

SQL> INSERT INTO Computer_Science VALUES('AJ00225', 'Swaroop', 'Second', 'PMCs');
1 row created.

SQL> INSERT INTO Computer_Science VALUES('AJ00296', 'Tejas', 'Second', 'BCA');
1 row created.

SQL> INSERT INTO Computer_Science VALUES('AJ00112', 'Geeta', 'First', 'BCA');
1 row created.

SQL> INSERT INTO Computer_Science VALUES('AJ00388', 'Rupa', 'First', 'PMCs');
1 row created.

SQL> select * from Computer_Science;

```
REGNO      NAME          YEAR          COMBINATION
-------------------------------------------------------------------------------
AJ00225    Swaroop       Second        PMCs
AJ00296    Tejas         Second        BCA
AJ00112    Geeta         First         BCA
AJ00388    Rupa          First         PMCs
```

1. Select all students from Physics and Computer Science

```
SQL> SELECT Regno, Name, Year, Combination FROM Physics
  2  UNION
  3  SELECT Regno, Name, Year, Combination FROM Computer_Science;

REGNO      NAME       YEAR          COMBINATION
-------------------------------------------------------------------------------
AJ00112    Geeta      First         BCA
AJ00225    Swaroop    Second        PMCs
AJ00296    Tejas      Second        BCA
AJ00325    Ashwin     First         PCM
AJ00385    Sarika     Third         PME
AJ00388    Rupa       First         PMCs

6 rows selected.
```

2. Select students common in Physics and Computer Science

```
SQL> SELECT Regno, Name, Year, Combination FROM Physics
  2  INTERSECT
  3  SELECT Regno, Name, Year, Combination FROM Computer_Science;

REGNO      NAME       YEAR          COMBINATION
-------------------------------------------------------------------------------
AJ00225    Swaroop    Second        PMCs
AJ00388    Rupa       First         PMCs
```

3. Display all student details those are studying in the second year

```
SQL> SELECT * FROM Physics WHERE Year = 'Second'
  2  UNION
  3  SELECT * FROM Computer_Science WHERE Year = 'Second';

REGNO      NAME       YEAR          COMBINATION
-------------------------------------------------------------------------------
AJ00225    Swaroop    Second        PMCs
AJ00296    Tejas      Second        BC
```

4. Display students who are studying both Physics and computer science in the second year

```
SQL> SELECT P.Regno,P.Name,P.Year,P.Combination
  2  FROM Physics P,Computer_Science CS
  3  WHERE P.Regno=CS.Regno AND P.Year='Second' AND CS.Year='Second';

REGNO      NAME       YEAR                 COMBINATION
-------------------------------------------------------------------------------
AJ00225    Swaroop    Second                  PMCs
```

5. Display the students studying only physics

```
SQL> SELECT * FROM Physics
  2  WHERE Regno NOT IN (SELECT Regno FROM Computer_Science);

REGNO      NAME      YEAR           COMBINATION
-------------------------------------------------------------------------------
AJ00325    Ashwin     First              PCM
AJ00385    Sarika     Third              PME
```

6. Display the students studying only Computer Science

```
SQL> SELECT * FROM Computer_Science
  2  WHERE Regno NOT IN (SELECT Regno FROM Physics);

REGNO        NAME           YEAR          COMBINATION
-------------------------------------------------------------------------
AJ00296      Tejas          Second            BCA
AJ00112      Geeta          First             BCA
```

7. Select all students having PMCs combination

```
SQL> SELECT Regno, Name, Year, Combination FROM Physics WHERE Combination =
'PMCs'
  2  UNION
  3  SELECT Regno, Name, Year, Combination FROM Computer_Science WHERE
Combination = 'PMCs';

REGNO      NAME       YEAR                 COMBINATION
-------------------------------------------------------------------------------
AJ00225    Swaroop    Second                  PMCs
AJ00388    Rupa       First                   PMCs
```

8. Select all students having a BCA combination

```
SQL> SELECT Regno, Name, Year, Combination FROM Computer_Science WHERE
Combination = 'BCA';

REGNO      NAME          YEAR          COMBINATION
-------------------------------------------------------------------------
AJ00296    Tejas         Second        BCA
AJ00112    Geeta         First         BCA
```

9. Select all students studying in the Third year

```
SQL> SELECT * FROM Physics WHERE Year = 'Third'
  2  UNION
  3  SELECT * FROM Computer_Science WHERE Year = 'Third';

REGNO    NAME    YEAR          COMBINATION
-------------------------------------------------------------------------------------
AJ00385  Sarika  Third         PME
```

10. Rename table Computer Science to CS

```
SQL> ALTER TABLE Computer_Science RENAME TO CS;

Table altered.
```

SQL> select * from CS;

```
REGNO    NAME     YEAR          COMBINATION
-------------------------------------------------------------------------------------
AJ00112  Geeta    First         BCA
AJ00225  Swaroop  Second        PMCs
AJ00296  Tejas    Second        BCA
AJ00325  Ashwin   First         PCM
AJ00385  Sarika   Third         PME
AJ00388  Rupa     First         PMCs
```